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MASTER THESIS

Learning to Couple:
Acute Effects of Rhythmic Stick Manipulation on Fractal Heart Rate Dynamics

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Abstract

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The beat-to-beat fluctuations of the healthy heart are correlated across time scales, and the short-term scaling exponent of detrended fluctuation analysis (DFA α_1) summarises this organisation in a single number that remains informative during physical activity. Movement practices built around handling an object are often described by practitioners as absorbing and restorative, but whether they leave a measurable acute trace in autonomic regulation has not been tested.

The present study recorded RR intervals with a Polar H10 chest strap while participants rested and while they performed two rhythmic stick-manipulation movements, isolations and spins, each for two minutes. α_1 was estimated per trial with an open analysis pipeline and compared between rest and movement, between the two movements, and between participants of differing expertise, with mean heart rate as a covariate to separate effects of skill from effects of physical load.

[todo: results in two to three sentences: direction of the shift, whether it survives the heart-rate adjustment, movement and expertise effects]

[todo: one to two sentences of interpretation and what follows for the stick as a research tool]

Keywords: heart rate variability, detrended fluctuation analysis, DFA alpha1, fractal physiology, object manipulation, motor skill, expertise, flow, Polar H10

Contents

1 Introduction and Experimental Objective **1**

2 Background **2**
2.1 Cardiac Autonomic Control and Heart Rate Variability 2
2.2 Fractal Physiology and Detrended Fluctuation Analysis 4
2.3 Acute Changes in DFA α_1 6
2.4 Rhythmic Movement, Sound and Cardiorespiratory Coupling 6
2.5 Skill, Expertise and Flow 8
2.6 Measuring RR Intervals Outside the Laboratory 10
2.7 Research Gap 10
2.8 Hypotheses 11

3 Methods **11**
3.1 Participants 11
3.2 The Stick and the Movements 12
3.3 Experimental Paradigm 12
3.4 Technical Setup 13
3.5 Real-time Metrics 14
3.6 Questionnaires 14

4 Offline Data Analysis **14**
4.1 Preprocessing 14
4.2 Heart Rate and Amplitude Indices 15
4.3 DFA α_1 15
4.4 Pilot Checks 16
4.5 Statistics 16

5 Results **17**
5.1 Data Quality and Sample 17
5.2 Descriptive Statistics 17
5.3 Task versus Rest (H1) 17
5.4 Isolations versus Spins (H2) 17
5.5 Expertise (H3) 17
5.6 Exploratory Analyses 17

6 Discussion **18**
6.1 Direction of the Shift 18
6.2 Skill or Load? 19
6.3 Expertise 19
6.4 The Stick as a Research Tool 20
6.5 Limitations and Future Research 20

7 Conclusion **20**

8 Appendix **26**

8.1 Abbreviations 26
8.2 Trial File Format 26
8.3 Supplementary Tables 26

1 Introduction and Experimental Objective

Even at rest, the interval between two consecutive heartbeats changes from one beat to the next. In healthy people these changes are correlated across time scales, so that the pattern of fluctuation over a few beats resembles the pattern over minutes and hours [19, 33]. This fractal organisation is thought to reflect the interplay of many regulatory loops acting on the sinoatrial node at once, and it degrades in characteristic ways with disease and ageing [19, 31]. The beat-to-beat timing of the heart therefore offers a continuous, non-invasive readout of how the nervous system is regulating the body.

Most research on heart rate variability (HRV) quantifies how much the heartbeat varies, using indices such as the root mean square of successive differences (RMSSD). These amplitude indices are informative at rest but collapse towards a floor as soon as physical activity rises, losing resolution exactly where a movement task places the participant [21]. The short-term scaling exponent of detrended fluctuation analysis (DFA α_1) quantifies how this variation is organised over windows of 4 to 16 beats. α_1 sits near 1.0 in rested, healthy adults, falls towards 0.75 around the first ventilatory threshold and towards 0.5 at high exercise intensity [34, 35]. Because it keeps a dynamic range under load and can be estimated from about two minutes of data [11, 34], it is a candidate marker for short bouts of physically light but attentionally demanding movement.

One such class of movement is skilled object manipulation. Juggling, poi, staff and stick work are widely practised, and practitioners describe them as absorbing, calming and restorative, often in the vocabulary of flow [13]. In cognitive neuroscience, object manipulation already serves as a model of skill learning: learning to juggle changes grey and white matter within weeks to months [17, 39]. The state of the body during manipulation itself has received far less attention. If a practice reliably induces a distinct state of engagement, that state should leave a trace in autonomic regulation during the practice itself, and α_1 offers a way to look for it.

A second motivation comes from the study of rhythmic coordination. During walking, running and cycling, breathing tends to lock into stable frequency ratios with the movement cycle, and an external rhythm such as a metronome can stabilise this locomotor–respiratory coupling and lower its energetic cost [2, 24]. Music and rhythmic sound, in turn, modulate arousal and can entrain cardiovascular rhythms [4, 44]. A rhythmic upper-limb movement therefore offers a plausible route from movement to breathing to the heart, and one in which mechanical coupling plays a smaller role than in locomotion. Whether such coordination behaves like a skill, weak in novices and stable in experts, is the broader question that gives this thesis its title.

The present study uses a stick of about 50 cm with a weight at each end, designed to produce a rhythmic figure-eight motion. Participants performed two families of movement with it. *Isolations* keep one point of the stick fixed in space while the hand moves around it; they are slow, precise and visually guided. *Spins* rotate the stick continuously around the hand; they are faster and more rhythmic. RR intervals were recorded with a Polar H10 chest strap and streamed to a laptop, where an open pipeline computed α_1 alongside heart rate and amplitude indices of HRV for each two-minute trial and for seated rest.

The experimental objective was to establish whether two minutes of stick manipulation shift α_1 relative to seated rest, and in which direction. Because α_1 falls with exercise intensity, a shift could

simply reflect physical load. A second objective was therefore to separate these two accounts by using mean heart rate as a covariate: an effect of manipulation counts as specific only if it exceeds what heart rate alone predicts. A third objective was to test whether the shift depends on the movement performed and on the participant's expertise with the stick, since the same visible movement can correspond to effortful, explicitly controlled action in a novice and to largely automatic action in an experienced practitioner [18].

Testing and validating the measurement pipeline itself constitutes a further objective. The capture system was built for this study from consumer hardware and open software, and the real-time α_1 reported by the streaming application can be compared directly with the estimate computed offline. Establishing how far these agree, and how robust α_1 is to the movement artifacts that object manipulation introduces, determines whether the stick can serve as a portable tool for studying states of skilled engagement outside the laboratory.

The thesis is organised as follows. Section 2 reviews cardiac autonomic control, the method and interpretation of DFA α_1, the evidence for acute changes in α_1, rhythmic coordination and its links to the heart, the psychophysiology of skill and flow, and the validity of chest-strap recording, and closes with the hypotheses. Section 3 describes participants, design and the capture system; Section 4 the offline signal processing and statistics. Sections 5 to 7 report, discuss and summarise the results.

2 Background

2.1 Cardiac Autonomic Control and Heart Rate Variability

2.1.1 The heart under central control

The sinoatrial node sets the pace of the heart, but its rate is continuously adjusted by the two branches of the autonomic nervous system. Sympathetic outflow accelerates the heart and increases contractility through noradrenergic signalling with a latency of several seconds. Parasympathetic outflow through the vagus nerve slows the heart through acetylcholine acting on muscarinic receptors, with a latency short enough to influence the very next beat [41]. Because the two branches act on different time scales, the beat-to-beat timing of the heart carries information about both: fast fluctuations are dominated by vagal activity, slower fluctuations reflect a mixture of vagal, sympathetic, baroreflex and thermoregulatory influences.

This outflow is organised centrally. Benarroch [3] described a *central autonomic network*, a set of reciprocally connected structures including the insular cortex, anterior cingulate and ventromedial prefrontal cortex, the amygdala, the hypothalamus, the periaqueductal grey, the parabrachial complex and the nucleus of the solitary tract, together with the medullary nuclei that generate sympathetic and vagal output. The network integrates visceral afferent information with descending signals related to emotion, attention and behaviour, and adjusts autonomic output accordingly. Its organisation has two properties that matter for the present thesis. First, it is distributed and hierarchical, so that autonomic output at any moment reflects the combined action of many loops. Second, its cortical components overlap with regions that support attention, error monitoring and self-regulation, so that cognitive and motor demands can shape autonomic output directly, in addition to the influence of metabolic need.

The ascending side of this circuitry is equally relevant. Interoceptive accounts place the posterior

and anterior insula at the centre of the brain's representation of bodily state, with a progression from primary representation of physiological signals to integrated, consciously accessible feelings in the anterior insula [12]. On this view the heart is both regulated by and represented in the brain, and the regulation of cardiac rhythm is one component of a broader loop linking bodily state, feeling and action.

2.1.2 Neurovisceral integration

The neurovisceral integration model makes the link between cardiac regulation and cognition explicit [42]. It proposes that prefrontal circuits supporting attention and self-regulation exert tonic inhibitory control over subcortical circuits involved in threat and defensive responses, and that the strength of this inhibition is expressed in the vagal control of the heart. High resting vagally mediated HRV is therefore taken to index a flexible regulatory system that can adjust quickly to changing demands, whereas low HRV indexes a system that is less able to disengage from a default defensive mode. A meta-analysis of neuroimaging studies supports the anatomical part of this account: HRV is associated with activity in the ventromedial prefrontal cortex and the amygdala, among other regions [43].

For cognitive neuroscience, the attraction of this framework is practical as well as theoretical. HRV can be recorded non-invasively and continuously during a task, at a temporal resolution of single heartbeats, with equipment that does not constrain movement. It therefore offers a window on how the brain regulates the body during activities, such as object manipulation, that are difficult to study with neuroimaging.

2.1.3 Amplitude indices of HRV and their limits

The standard indices of HRV quantify the amount of variability. In the time domain, the standard deviation of normal-to-normal intervals (SDNN) captures overall variability, and the root mean square of successive differences (RMSSD) and the percentage of successive differences larger than 50 ms (pNN50) capture beat-to-beat, predominantly vagal, variability [41]. In the frequency domain, power in the high-frequency band (0.15–0.40 Hz) tracks respiratory sinus arrhythmia and vagal modulation, while low-frequency power (0.04–0.15 Hz) reflects a mixture of influences including the baroreflex [41]. Recommendations for psychophysiological research specify recording conditions, minimum durations and the control of confounds such as caffeine, recent exercise, posture and breathing [30].

Amplitude indices have two limitations for the present question. First, they are strongly intensity-dependent: vagal withdrawal at the onset of physical activity drives RMSSD and high-frequency power towards a floor, so that they lose resolution as soon as a task involves movement [21]. Second, the amount of variability does not distinguish a heartbeat that varies richly across time scales from one that varies in a simple, repetitive way. A strongly periodic signal, such as the large oscillations produced by slow paced breathing, yields high RMSSD and high spectral power but little complexity. Reflex tests illustrate the point from the opposite direction: resting RMSSD shares only about half or less of its variance with other markers of cardiovagal function, such as the response to paced deep breathing or to active standing [6]. No single amplitude index captures autonomic regulation in full.

2.2 Fractal Physiology and Detrended Fluctuation Analysis

2.2.1 Complexity and health

Fractal physiology addresses the second limitation. Many physiological signals, including the heartbeat, respiration and the stride intervals of human gait, show fluctuations that are statistically self-similar across time scales: their variability over short windows resembles their variability over long windows when appropriately rescaled [19, 22]. In the heartbeat of healthy young adults, these fluctuations show long-range correlations close to $1/f$ behaviour, the boundary between uncorrelated noise and strongly correlated, Brownian-like drift [19, 33].

The key observation of this literature is that disease and ageing erode fractal organisation, and that they do so in two directions [19, 31]. Some conditions push the heartbeat towards uncorrelated randomness, in which each interval carries little information about the next. Others push it towards excessive regularity, in which a single dominant rhythm takes over, as in the strongly periodic heart rate oscillations of severe heart failure. In both cases the system loses the capacity to respond across many time scales at once. On this view, physiological health corresponds to adaptable organisation across time scales, and a loss of complexity can be detected as a departure from the healthy scaling regime in either direction. Short-term fractal scaling of RR intervals has clinical relevance: in patients with depressed left ventricular function after myocardial infarction, a reduced short-term scaling exponent predicted mortality more strongly than traditional HRV indices [26].

This framework connects naturally to the notion of biological robustness. Kitano [29] characterised robustness as a fundamental property of complex biological systems: the capacity to maintain function in the face of perturbations, achieved through feedback control, modularity, redundancy and decoupling. A heartbeat regulated by many interacting loops operating on different time scales is, on this reading, the observable signature of a robust control architecture, and fractal scaling is one way of quantifying it. Network physiology extends the same idea to exercise, treating the organism as a network of interacting systems whose coupling reorganises with physical and cognitive demands [1].

2.2.2 Computing DFA

Detrended fluctuation analysis was introduced to quantify correlations in non-stationary time series such as heartbeat recordings [33]. For a series of N RR intervals RR_i with mean $\overline{RR}$, the series is first integrated,

$$y(k) = \sum_{i=1}^{k} \left(RR_i - \overline{RR} \right), \qquad k = 1, \ldots, N. \tag{1}$$

The integrated series is divided into non-overlapping boxes of length n beats. In each box a least-squares line $y_n(k)$ is fitted and subtracted, which removes local trends. The root-mean-square fluctuation of the detrended series,

$$F(n) = \sqrt{\frac{1}{N} \sum_{k=1}^{N} \left[y(k) - y_n(k) \right]^2}, \tag{2}$$

is computed for a range of box sizes. If the series is fractal, $F(n)$ grows as a power of n, $F(n) \propto n^{\alpha}$, and the scaling exponent α is the slope of $\log F(n)$ against $\log n$. Heartbeat series typically show a

crossover between short and long time scales, so two exponents are estimated: α_1 over box sizes of 4 to 16 beats and α_2 over 16 to 64 beats [33]. The present thesis concerns α_1, which can be estimated from short recordings and changes within minutes.

2.2.3 Interpreting α_1

The value of α maps onto classes of signal. An uncorrelated series (white noise) gives $\alpha = 0.5$. A series with $1/f$ correlations gives $\alpha = 1.0$. Brownian motion, the integral of white noise, gives $\alpha = 1.5$. Values between 0.5 and 1.0 indicate persistent but stationary correlations, and values between 1.0 and 1.5 indicate increasingly non-stationary, drift-dominated behaviour. Hoos and Gronwald [25] argue that these values should be read against the distinction between fractional Gaussian noise (fGN, $\alpha < 1$) and fractional Brownian motion (fBM, $\alpha > 1$), with the healthy resting heartbeat sitting near the boundary between the two. On this reading, $\alpha_1 \approx 0.75$ marks the midpoint of the fGN range, and $\alpha_1 \approx 0.5$ marks the loss of short-term correlation. Table 1 summarises the landmarks used throughout this thesis.

Table 1: Landmark values of the short-term scaling exponent α_1.

α_1	Signal class	Typical physiological context
≈ 1.5	Brownian-like, strongly correlated drift	Excessive regularity, e.g. slow periodic oscillations dominating the series [19]
≈ 1.0	$1/f$, boundary of fGN and fBM	Healthy rest and very low exercise intensity [21]
≈ 0.75	Midpoint of the fGN range	First ventilatory threshold in incremental exercise [35]
≈ 0.5	Uncorrelated (white noise)	High exercise intensity; loss of organisation [20]

2.2.4 Window length, detrending and reference settings

Because α_1 spans only 4 to 16 beats, it can be estimated from short recordings, but the estimate becomes unstable when too few boxes are available. In exercise research, α_1 is commonly computed over a two-minute window, recomputed every few seconds as the window slides along the recording [34, 35]. The two-minute convention is supported by analyses of how data length and non-stationarity affect DFA estimates [11] [verify: exact claim in Chen et al. on minimum length], and it corresponds to roughly 120 to 200 beats at the heart rates seen in light activity. The present study adopts the two-minute window because it is also close to the longest duration for which a skilled manipulation movement can be sustained without breaking off.

Two further settings affect the estimate. The first is detrending of the RR series before DFA. Reference implementations, including the widely used Kubios software, apply smoothness-priors detrending [40] with a smoothing parameter of $\lambda = 500$ to remove very slow trends that would otherwise inflate the fluctuation function [34]. The second is the treatment of artifacts. Missed, extra and ectopic beats distort α_1 more strongly than amplitude indices, and artifact rates above a few percent bias the estimate [34]. Hoos and Gronwald [25] argue that these settings, together with window length and box-size range, need to be standardised and reported explicitly before α_1 values can be compared between studies. Section 4 specifies the settings used here.

2.3 Acute Changes in DFA α_1

2.3.1 Rest versus activity

The heartbeat's correlation structure changes between rest and activity. Karasik et al. [27] reported that correlations in heartbeat fluctuations differ between rest and exercise, with the short-range structure changing more than the long-range structure [verify: direction and scale of the effect in Karasik et al.]. Subsequent exercise research established that α_1 decreases systematically with increasing intensity. A systematic review found this pattern consistently across endurance exercise studies: α_1 sits around 1.0 at rest and low intensity, falls through the moderate range and approaches 0.5 or below at high intensity [20]. The decline is generally interpreted as a loss of correlation as the multiple regulatory influences on the heart are progressively replaced by a dominant drive from central command and metabolic feedback [1, 21].

2.3.2 α_1 as an intensity marker

The intensity dependence of α_1 has made it a candidate biomarker for exercise prescription. Rogers et al. [35] proposed that the heart rate at which α_1 crosses 0.75 during an incremental test approximates the first ventilatory threshold, the boundary between low and moderate intensity. Later work broadened this to a general account of α_1 as a marker of intensity distribution in training [21, 34]. The proposal has been challenged. Cassirame et al. [9] examined DFA-derived thresholds in an incremental cycling test and questioned how reliably they correspond to physiological thresholds at the individual level [verify: exact findings of Cassirame et al.], and Hoos and Gronwald [25] cautioned that the interpretation of fixed cut-offs depends on reference settings that are not yet standardised. For the present thesis, the relevant point is that α_1 responds within minutes to changes in physiological state.

2.3.3 Beyond intensity

A smaller body of work shows that α_1 also responds to factors other than momentary intensity. In ultramarathon runners, α_1 measured at the same low running intensity was lower after the race than before it, suggesting that fatigue alters the organisation of heart rate regulation independently of workload [36]. Short-term DFA has also been applied to psychological stress paradigms [todo: find and add a verified source on short-term DFA under cognitive or psychological stress]. These findings matter for the present question because they show that two conditions with the same heart rate can differ in α_1. This is the logic on which the heart-rate adjustment in the present design rests.

What is missing from this literature is a task in which physical intensity is low but attentional and motor demands are high. Almost all acute α_1 studies concern endurance exercise, in which intensity and regulatory demand rise together. Object manipulation decouples them: it raises heart rate only modestly but places continuous demands on visuomotor prediction and fine force control.

2.4 Rhythmic Movement, Sound and Cardiorespiratory Coupling

2.4.1 Locomotor–respiratory coupling

Rhythmic movement interacts with the body's other rhythms. During walking, running, cycling, rowing and wheelchair propulsion, breathing tends to lock into stable frequency ratios with the movement cycle, a phenomenon called locomotor–respiratory coupling (LRC) [5, 7]. In quadrupeds,

coupling is enforced mechanically by the loading of the thorax with each stride [7]; in humans it is weaker and more variable, and its strength differs between individuals and tasks [5]. Bardy et al. [2] distinguish three sources of coupling: mechanical interactions, such as the displacement of the viscera with each step; neurophysiological interactions, such as afferent signals from moving limbs reaching respiratory centres in the brainstem; and informational interactions, such as external rhythms the person attends to.

LRC can be modelled as the interaction of two coupled nonlinear oscillators. Hoffmann et al. [24] used the sine circle map, in which the ratio of the two frequencies (the winding number) and the strength of the coupling determine whether the system settles into a stable mode-locked ratio. The stable ratios are ordered in a Farey tree: low-order ratios such as 1:2 occupy wider regions of stability than high-order ratios, and are therefore more resistant to perturbation [23, 24].

2.4.2 Sound-induced stabilisation

External sound strengthens this coupling. In a cycling study, participants synchronised either their pedalling or their breathing with a metronome set to their own preferred frequency [24]. Auditory pacing stabilised LRC and was accompanied by a significant reduction in energy expenditure. The effect was asymmetrical: pacing the locomotor rhythm had a stronger stabilising influence than pacing the respiratory rhythm, and when participants cycled in time with the tone their breathing frequency shifted towards more stable frequency ratios, as the sine circle map predicts. A follow-up study varied the frequency of pacing [23]. Sound stabilised LRC at all frequencies tested and for both paced systems, but the effect was strongest when the pacing frequency matched the participant's preferred rhythm, and frequencies far from the preferred ones led to a loss of stability. The authors concluded that coupling runs in both directions, from movement to breathing and from breathing to movement.

Bardy et al. [2] reviewed both studies and drew out the dependence on starting state: sound stabilised participants whose coupling was initially unstable but could destabilise those whose coupling was already stable [verify: whether this is reported in Bardy et al. 2015 or Hoffmann et al. 2012]. The effect of rhythmic information therefore depends on how well coordinated the person already is, which is precisely the kind of individual difference that expertise might produce.

2.4.3 Music, rhythm and the heart

The cardiovascular effects of music have been studied for decades, with results that range from careful laboratory physiology to broad clinical claims. Narrative reviews conclude that music can modulate heart rate, blood pressure, HRV and stress hormones, but that effects depend on the music and the listener and that the underlying mechanisms remain poorly understood [10, 44]. Some of these reviews go further, recommending particular composers and warning against particular genres [44]; such claims rest on limited evidence and are noted here as context only. A randomised controlled study gives a more measured picture: 25 minutes of music by Mozart or Strauss lowered blood pressure and heart rate relative to the values before listening, whereas music by ABBA did not, and serum cortisol decreased in all groups, including a control group resting in silence [45].

Laboratory work on tempo is more directly relevant. Faster music elicits greater sympathetic arousal, measured as skin conductance, than slower music, and the frequency of skin conductance responses is higher during music than during silence, consistent with musical beats acting as structural features

that capture attention [16]. Bernardi et al. [4] recorded heart rate, respiration, blood pressure, cerebral blood flow velocity and skin vasomotion while 24 young adults, half of them trained choristers, listened to music with different profiles. Crescendos produced proportional increases in blood pressure and skin vasoconstriction, uniform emphasis produced the reverse, and musical phrases with a period of about ten seconds, close to the 0.1 Hz rhythm of blood-pressure Mayer waves, entrained cardiovascular autonomic variables. Notably, the responses did not differ qualitatively between musicians and non-musicians. A recent review in the Cambridge Elements series integrates such findings into an account of how auditory rhythm interacts with cardiac, respiratory and motor rhythms in perception and action [47]. Wearable sensing now allows these relations to be measured outside the laboratory: an earable system predicted the direction of physiological change during music listening from vital signs alone in about 70% of cases [8].

Rhythmic activity shared between people can couple their hearts as well. When pairs of non-experts vocalised together, their HRV became more coupled during long, synchronous notes than during short notes or at baseline [37]. Most of this coupling was attributable to respiratory sinus arrhythmia, but part of it persisted after the influence of breathing was removed, and the frequency of vocal onsets matched the frequency at which coupling peaked. Coupling did not, however, map clearly onto participants' subjective sense of togetherness.

2.4.4 Implications for an upper-limb rhythmic task

Three conclusions follow for the present study. First, rhythmic movement and rhythmic sound both have documented routes to cardiorespiratory rhythms, mainly via breathing and respiratory sinus arrhythmia, with some coupling remaining once breathing is accounted for [37]. Second, the stick's figure-eight motion is a rhythmic upper-limb movement without the impact loading of locomotion, so any coupling it produces is likely to be mainly neurophysiological and informational [2]. Third, the effect of rhythmic information depends on the starting state of the person, which motivates measuring expertise. The present study records cardiac dynamics only, without a breathing signal or a metronome, and asks whether a downstream signature of coupling is visible there. Coupling serves as one candidate mechanism among several, alongside effort, attention and arousal, and the discussion returns to how it could be isolated in future work.

2.5 Skill, Expertise and Flow

2.5.1 Object manipulation as a model of skill

Object manipulation has a place in cognitive neuroscience as a model of skill learning. Three months of juggling practice produced transient increases in grey matter in the mid-temporal visual motion area hMT/V5 and the left posterior intraparietal sulcus, which partly reversed when practice stopped [17]. Six weeks of juggling training changed white-matter architecture beneath the right posterior intraparietal sulcus [39]. These studies established juggling as a model of experience-dependent plasticity. Their focus, however, is structural and long-term. They say little about the state of the person while manipulating an object, which is the focus here.

Classical accounts describe skill acquisition in three stages [18]. In the *cognitive* stage, the learner works out what to do; movement is effortful, error-prone and under explicit attentional control. In the *associative* stage, errors fall and movement becomes more consistent as the learner refines

the mapping between intention and outcome. In the *autonomous* stage, movement runs with little conscious control, freeing attention for other demands. A stick-naive participant performing an isolation for the first time is in the cognitive stage; an experienced practitioner performing the same movement is close to autonomy. The same visible movement can therefore correspond to very different internal states, and any physiological measurement of practice has to take expertise into account.

The two movements in this study differ along the same lines. Isolations demand that one point of the stick stay fixed in space while the hand travels around it, which requires continuous visual feedback and fine corrections. Spins are faster and, once learned, more self-sustaining: the rotation carries momentum, and the hand's role shifts from placing the stick to maintaining a rhythm. Isolations thus load prediction and control, spins load rhythm and timing. Expertise is expected to change both, possibly in different ways.

2.5.2 The psychophysiology of flow

Flow denotes a state of complete absorption in an activity whose challenge matches one's skill, accompanied by a merging of action and awareness, a sense of control and a loss of self-consciousness [13]. Its physiology has mostly been studied with amplitude measures. In expert pianists playing pieces they knew well, flow was associated with higher heart rate, deeper breathing and activity of the zygomaticus major muscle, a pattern the authors interpreted as positive affect combined with effortful attention [14] [verify: exact physiological findings in de Manzano et al.]. In a task in which demands were matched to participants' skill, the matched condition was associated with lower HRV and higher salivary cortisol than conditions of boredom or overload [28] [verify: direction of HRV and cortisol effects in Keller et al.]. Peifer et al. [32] brought these findings together by showing that flow relates to sympathetic arousal in an inverted U and to parasympathetic activity positively, suggesting co-activation of both autonomic branches during flow.

Neurocognitive accounts point in a compatible direction. The transient hypofrontality hypothesis proposes that flow involves a temporary down-regulation of explicit, prefrontally mediated control, allowing implicit, well-practised processes to run with less interference [15]. Imaging of experimentally induced flow, using a task adaptively matched to each participant's skill, found reduced activity in the medial prefrontal cortex and amygdala and increased activity in the inferior frontal gyrus and putamen [46]. The medial prefrontal cortex and amygdala are precisely the regions implicated in neurovisceral integration (Section 2.1), which gives a plausible anatomical route from a flow-like state to cardiac regulation.

2.5.3 The case for a complexity marker

Correlation structure may capture co-activation of competing autonomic inputs, a pattern that amplitude indices summarise poorly. If a state of skilled absorption involves several regulatory loops operating together, its signature should appear in how heart rate variation is organised, which is what α_1 quantifies. Inducing flow is outside the primary aim of the present study, and participants include novices for whom the task is unlikely to produce it. Flow nevertheless motivates the choice of α_1 as the outcome and the inclusion of expertise as a factor: if the organisation of cardiac regulation during manipulation changes with skill, experts are where a flow-related signature would first be expected.

2.6 Measuring RR Intervals Outside the Laboratory

Electrocardiography (ECG) is the reference method for recording RR intervals, but chest-strap sensors are now widely used in research because they are cheap, portable and tolerate movement. The Polar H10 records a single-lead ECG through electrodes in the strap and transmits RR intervals at a resolution of 1/1024 s over Bluetooth Low Energy.

Two validation studies frame its use here. Schaffarczyk et al. [38] compared the H10 with simultaneous ECG in 25 recreational athletes during rest and an incremental cycling test to exhaustion. Agreement was high at rest before and after exercise (e.g. $r = 0.95$ and ICC $= 0.95$ for α_1 before exercise) and correlations remained above 0.93 during exercise. During exercise, however, all variables showed a statistically significant bias. For RR intervals and heart rate the bias and limits of agreement were negligible, but for α_1 they widened with intensity, from a bias of 0.9–8.6% with limits of agreement of +11.6 to −9.9% at low intensity to limits of +58.1 to −40.9% at high intensity. α_1 values in the uncorrelated range, below about 0.5, were the least reliable. Blalock et al. [6] compared the H10 with ECG in 43 healthy young adults during supine rest, paced deep breathing and active standing. Agreement for mean RR interval, RMSSD and SDNN, and for standard cardiac reflex tests, was excellent (concordance correlation ≥ 0.99, mean absolute percentage error below 1%, 95% limits of agreement within about ± 1.7 ms), and the authors concluded that the H10 is effectively interchangeable with laboratory ECG for these measures.

Two implications follow. For RR intervals and amplitude indices at rest and during light activity, the H10 can be treated as equivalent to ECG. For α_1, it is valid at rest and at low intensity, which is the range expected during stick manipulation, but its error grows with intensity and with movement of the upper body, which object manipulation introduces. Artifact rates must therefore be reported per trial, and agreement between the real-time and offline α_1 estimates is itself a question for the present study.

2.7 Research Gap

The review identifies three gaps. The first concerns the *domain*: research on acute changes in α_1 concerns endurance exercise, in which physical intensity and regulatory demand rise together; brief, attention-demanding motor skills at low intensity have not been studied. The second concerns the *index*: the psychophysiology of flow and skilled engagement relies on amplitude indices such as RMSSD and high-frequency power, and the organisation of variability during skilled action has not been examined. The third concerns *expertise*: research on rhythmic coupling shows that the effect of rhythmic information depends on the person's starting state, and research on skill shows that the same movement corresponds to different internal states in novices and experts, but no study has asked whether the cardiac signature of a movement depends on expertise with it.

Underlying all three is a methodological problem. Because α_1 falls with exercise intensity, any change during a movement task could reflect physical load alone. A study of skilled manipulation must therefore treat heart rate as a covariate, so that an effect counts as specific only if it exceeds what heart rate alone predicts.

2.8 Hypotheses

Young, healthy participants are likely to rest near $\alpha_1 \approx 1.0$, which leaves little room for an *increase* to be interpreted as an improvement. The hypotheses are therefore formulated in terms of direction and specificity, and interpreted with the two-directional complexity framework (Section 2.2): a shift towards 0.5 indicates a loss of short-term correlation, a shift towards 1.5 indicates increasing regularity, and stability near 1.0 indicates preserved organisation under task demand. All hypotheses were specified before the main sample was collected [confirm: pre-registration with Timo, date and platform].

H1 (primary): Task versus rest.
Mean α_1 during stick manipulation differs from α_1 during seated rest (two-sided test). The direction of the shift, and the position of task values relative to the landmarks 0.5, 0.75 and 1.0, is reported regardless of significance.

H2 (secondary): Movement type, adjusted for heart rate.
The shift in α_1 differs between isolations and spins after adjusting for mean heart rate. Spins are faster and more rhythmic, so their shift is expected to be largely explained by heart rate; isolations are slow, precise and visually guided, so any shift beyond what heart rate explains would point to regulatory engagement.

H3 (secondary): Expertise.
Expertise moderates the shift. Novices, operating in the cognitive stage of skill, are expected to show a larger absolute departure from their resting α_1 than experienced practitioners at comparable heart rate, because the same movement demands more explicit control from them. Expertise is analysed both as a group factor and, where the sample allows, as a continuous measure of practice [confirm: expertise grouping and measure].

Three questions are examined without a directional hypothesis: whether α_1 at a second rest period after the movements differs from the first (carry-over); how closely the real-time α_1 reported by the streaming application agrees with the offline estimate; and whether the dominant and non-dominant hand differ when the same movement is performed with each.

3 Methods

3.1 Participants

Data were collected in two phases. In the first phase (October 2026), five participants completed the full protocol; these recordings also served to check data quality and to fix the remaining protocol parameters. In the second phase (November and December 2026), about ten further participants were recruited with the aim of covering a range of expertise with the two movements, from stick-naive adults to practitioners with regular practice [confirm: final N and recruitment channels]. The target sample was therefore $N \approx 15$.

Inclusion criteria were age between 18 and 45 years, no known cardiovascular, respiratory or neurological condition, and the ability to perform both movements with each hand for two minutes. Stick-naive participants received a short standardised instruction and practice period before recording

(Section 3.3). Exclusion criteria were arrhythmia, an implanted cardiac device, pregnancy, and medication known to affect heart rate or HRV, such as beta-blockers. Participants were asked to avoid caffeine for two hours and strenuous exercise for 24 hours before the session, and to avoid a heavy meal in the preceding two hours [30]. Compliance with these requests, hours of sleep the previous night and handedness were recorded on arrival.

Expertise was assessed with a short questionnaire covering years of practice with the stick or comparable flow tools (staff, poi, juggling, contact props), typical practice hours per week, and a self-rating of proficiency for each movement on a scale from 1 to 10 [confirm: expertise items and grouping rule]. Participants were assigned to a novice or an experienced group by a rule fixed before recruitment of the second phase, and the continuous measures were retained for a secondary analysis (Section 4.5).

All participants gave written informed consent. [todo: ethics: name the approving committee or state that the study falls under the unit's existing approval; add reference number]

With a paired design, $N = 15$ provides 80% power to detect a within-participant effect of $d \approx 0.78$ at $\alpha = .05$ (two-sided). Effects smaller than this, and all between-group comparisons of expertise, are therefore reported as estimates with confidence intervals.

3.2 The Stick and the Movements

The stick is about 50 cm long with a weight at each end [confirm: exact length, mass and material]. The weighting moves the centre of mass outwards and gives the stick rotational inertia, so that once set in motion it tends to continue in a figure-eight path around the hand. The design was developed within a movement practice that treats structured variability as the basis of skilled coordination: movement that keeps a recognisable pattern while varying from cycle to cycle [confirm: whether to cite the EightOS / seks.design material here].

Two families of movement were recorded. In an *isolation*, one point of the stick, typically its centre or one end, is held fixed in space while the hand and the rest of the stick move around it. Isolations are slow, demand continuous visual monitoring and fine force control, and break down visibly when control lapses. In a *spin*, the stick rotates continuously around the hand or fingers in a figure-eight. Spins are faster, rely mainly on timing and momentum, and produce a regular rhythm once established. Each movement was performed with the left and with the right hand. Participants were instructed to keep the movement going for the whole trial at a pace they could sustain, to restart immediately after a drop, and to breathe naturally. Drops were counted by the experimenter.

3.3 Experimental Paradigm

The study used a within-participant design with one factor, *condition*, at three levels (seated rest, isolation, spin), crossed with the between-participant factor *expertise* (novice, experienced). Hand (left, right) was recorded within each movement condition and treated as a secondary factor. Seated rest without the stick served as the control condition; each participant was their own control.

Each session followed the sequence in Table 2. After consent, questionnaires and fitting of the chest strap, participants sat quietly for a pre-task rest period. The four movement trials (isolation left, isolation right, spin left, spin right) followed in an order counterbalanced across participants [confirm:

counterbalancing scheme: Latin square vs randomised]. Each movement trial was followed by seated recovery, so that every trial started from a comparable resting state. A post-task rest period closed the session.

Table 2: Session timeline. Durations in square brackets are still to be fixed from the first-phase data.

Step	Duration	Content
1	10 min	Consent, questionnaires (demographics, handedness, expertise, caffeine, sleep), strap fitting, signal check
2	5 min	Familiarisation: practice of both movements with each hand (novices); warm-up (experienced)
3	[2–5 min]	Pre-task rest, seated, eyes open, no stick
4	4 × 2 min	Movement trials (isolation and spin, each hand), counterbalanced order
	[2 min] each	Seated recovery after each movement trial
5	[2–5 min]	Post-task rest, seated
6	5 min	Post-session ratings and debrief
	≈ 45 min	Total

The trial duration of two minutes was chosen as a compromise between two constraints. α_1 is conventionally estimated from a two-minute window (Section 2.2), so shorter trials would not yield a stable estimate. Longer trials, in turn, exceed the duration for which isolations in particular can be sustained without fatigue of the forearm and repeated drops, which would turn the trial into a series of restarts. [confirm: 2 min vs 3 min: if trials are extended to 3 min, α_1 is taken from the final 2 min so that the onset transient is excluded]

Rest periods were recorded with the same capture system and the same durations as the analysis windows, so that rest and movement estimates were based on comparable amounts of data. The pre-task rest is the baseline for H1; the post-task rest serves the carry-over analysis.

3.4 Technical Setup

Figure 1 gives an overview of the recording chain. RR intervals were recorded with a Polar H10 chest strap (Polar Electro, Kempele, Finland) worn at the level of the lower sternum with moistened electrodes. The H10 transmits RR intervals at a resolution of 1/1024 s through the standard Bluetooth Low Energy Heart Rate Measurement characteristic, in which a single notification can carry several RR values.

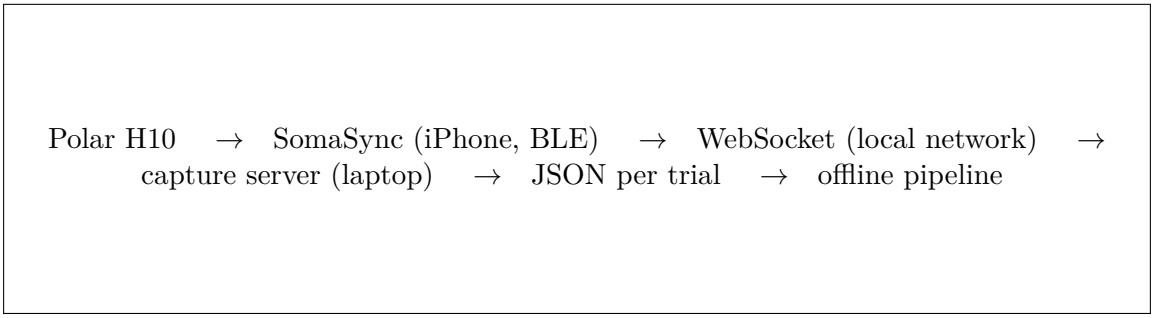


Figure 1: Recording chain. [todo: replace with a drawn diagram and a photo of the setup]

A first version of the setup connected the H10 directly to a browser page through the Web Bluetooth

interface. On the laptop used for recording, this connection repeatedly dropped within seconds of being established, so the final setup routed the signal through a phone. The H10 was paired with the SomaSync application on an iPhone [confirm: app version], which maintains the Bluetooth connection natively and streams each incoming RR interval, together with its own real-time HRV metrics, as JSON messages over a WebSocket connection on the local network.

A custom capture server (Node.js) running on the laptop received this stream. Its web interface, opened in a browser on the same laptop, allowed the experimenter to enter the participant identifier, the condition label and the trial duration before starting each trial. The server stamped each received RR interval with the time of arrival, kept a running count of samples and stopped the trial automatically when the set duration had elapsed, so that every trial had the same length without manual timing. Each trial was saved as a JSON file named from the condition, participant and start time, containing the list of RR intervals with timestamps, the trial metadata and the target duration, together with a raw log of every message received. Keeping the raw log means that any value reported by the streaming application can be recomputed or checked later. The code for the capture server and the analysis pipeline is available in the project repository [todo: repository link or archive DOI].

3.5 Real-time Metrics

In addition to the raw RR intervals, SomaSync reports its own running estimates of heart rate, RMSSD, SDNN, pNN50 and α_1, which were stored with each trial. The settings behind these estimates (window length, detrending, artifact handling) are not documented in enough detail to be reproduced [confirm: check with SomaSync documentation or developer]. They were therefore not used for the primary analyses. Instead, they were compared with the offline estimates computed from the same RR intervals (Section 4.5), which tests whether a consumer application can stand in for an offline pipeline in future, larger studies.

3.6 Questionnaires

Before the session, participants completed questions on demographics, handedness, expertise (Section 3.1), caffeine intake, sleep and recent exercise. After each movement trial, participants rated perceived difficulty and perceived exertion [confirm: scales: e.g. Borg CR10 for exertion], and after the session they completed a short flow questionnaire for each movement [confirm: instrument, e.g. Flow Short Scale]. These ratings are used descriptively and in exploratory analyses only.

4 Offline Data Analysis

All offline analyses were run in Python (NumPy, SciPy, statsmodels) with a pipeline written for this study. The pipeline reads each trial file, corrects artifacts, computes HRV metrics and α_1, and stores the results in a local database from which a browser dashboard displays each trial alongside the metrics reported in real time by the streaming application [todo: versions of Python and packages].

4.1 Preprocessing

RR intervals were taken in the order received and converted to milliseconds. Artifacts were identified in two steps. First, intervals outside 300–2000 ms, corresponding to heart rates outside about

30–200 beats per minute, were marked as physiologically implausible. Second, each remaining interval was compared with the last accepted interval, and intervals differing from it by more than 20% were marked as artifacts. This percentage rule catches missed beats, extra beats and most ectopic beats. Marked intervals were replaced by linear interpolation between the neighbouring accepted intervals, which keeps the series continuous and preserves its length, as DFA requires.

The proportion of corrected intervals was computed for every trial and is reported in Section 5. Because α_1 is more sensitive to artifacts than amplitude indices, trials with more than 5% corrected intervals were excluded from the analyses of α_1 [confirm: threshold: 5% vs 3%]; bias in α_1 has been reported to become relevant above a few percent [34]. A sensitivity analysis repeated the main models with a stricter threshold of 3%.

4.2 Heart Rate and Amplitude Indices

For each trial the pipeline computed mean heart rate as $60\,000/\overline{RR}$ (beats per minute), SDNN as the standard deviation of the corrected RR intervals, RMSSD as

$$\mathrm{RMSSD} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N-1}(RR_{i+1} - RR_i)^2}, \tag{3}$$

and pNN50 as the percentage of successive differences larger than 50 ms [41]. These indices are reported alongside α_1 to allow comparison with earlier work on flow and skilled performance, which has relied on them, and mean heart rate enters the models as the covariate that indexes physical load.

4.3 DFA α_1

α_1 was computed following Peng et al. [33] as described in Section 2.2. The corrected RR series was integrated after subtraction of its mean and divided into non-overlapping boxes for each box size $n = 4, 5, \ldots, 16$ beats. In each box a linear trend was fitted by least squares and removed, and the fluctuation function $F(n)$ was computed as the root mean square of the residuals across all boxes. α_1 was the slope of a least-squares line fitted to $\log F(n)$ against $\log n$. Estimates were not computed for series shorter than 64 beats, which ensures at least four boxes at the largest box size.

The primary outcome was a single α_1 value per trial, computed over the full two-minute trial. To describe how α_1 evolved within longer recordings, such as the rest periods, the pipeline also computed a sliding-window estimate over windows of 120 beats moved in steps of 30 beats. These within-trial traces are used descriptively only: a two-minute trial contains essentially one independent estimate, so within-trial dynamics cannot be tested.

Two settings differ from the reference implementation used in most exercise studies and were aligned with it before the final analysis [confirm: implemented and re-run]. First, smoothness-priors detrending with $\lambda = 500$ [40] was applied to the RR series before DFA, to remove slow trends in heart rate that would otherwise inflate the fluctuation function at the larger box sizes. Second, the sliding-window traces were defined in time (two-minute windows, recomputed every 5 s), so that windows contain comparable durations of data across conditions with different heart rates [34]. Results with both the original and the aligned settings are reported, so that the effect of these choices is visible [25].

4.4 Pilot Checks

The first recordings were used to check the pipeline end to end. They confirmed that trials started and stopped at the set duration, that RR intervals arrived without gaps over the WebSocket link, and that the dashboard reproduced the files on disk. They also showed that the real-time α_1 reported by the streaming application could differ noticeably from the offline estimate for the same trial; in one rest recording, for example, the two values were 1.21 and 1.48. Differences of this size are large relative to the shifts expected between conditions, which is why the primary analyses use the offline estimate only and the agreement between the two is analysed formally (Section 4.5). [todo: summarise artifact rates and heart rates from the first five participants once processed]

4.5 Statistics

Table 3 summarises the analysis plan. All models were linear mixed-effects models with a random intercept per participant, fitted by restricted maximum likelihood; fixed effects were tested with Wald tests and reported with 95% confidence intervals. Movement trials were averaged across hands for H1 and H3 and entered separately for H2, with hand as a covariate. Model assumptions were checked with residual plots; where residuals departed clearly from normality, results were checked with a non-parametric test on per-participant means.

Table 3: Analysis plan. $\Delta\alpha_1$ is the α_1 of a movement trial minus the participant's pre-task rest value.

Question	Outcome	Model
H1: task vs rest	α_1	condition (rest, isolation, spin) + random intercept; paired t-test on per-participant means as check
Direction of shift	$\Delta\alpha_1$	sign test on the number of participants whose α_1 rises vs falls; position of task values relative to 0.5, 0.75, 1.0
H2: isolation vs spin	α_1	movement + mean HR + hand + random intercept (movement trials only)
H3: expertise	$\lvert\Delta\alpha_1\rvert$	expertise group + ΔHR + movement + random intercept; secondary: expertise as continuous measure
Carry-over	α_1	post-task vs pre-task rest, paired
Hand	α_1	dominant vs non-dominant hand, coded by handedness (exploratory)
Agreement	α_1	Bland–Altman analysis, real-time vs offline estimate

For H1, the contrast of interest was rest versus movement, with the two movement levels compared with rest separately and jointly. Because no direction was predicted, the direction of the shift was reported for every participant, and the proportion of increases and decreases was tested with a sign test. For H2, adding mean heart rate as a covariate tests whether a difference between isolations and spins remains once their difference in physical load is accounted for; the analysis was repeated with the change in heart rate from rest as the covariate. For H3, the absolute change in α_1 was used because the direction of the shift was not predicted, while the heart-rate change entered as a covariate so that a larger departure in novices could not simply reflect greater effort.

H2 and H3 were corrected for multiple comparisons with the Holm procedure. Exploratory analyses (carry-over, hand, agreement, and relations with questionnaire ratings) were not corrected and are reported as estimates with 95% confidence intervals. Agreement between real-time and offline α_1 was assessed with Bland–Altman analysis, reporting the mean difference (bias) and 95% limits of

agreement, and checking whether the difference depended on the magnitude of α_1, as reported for the H10 against ECG [38].

5 Results

[todo: write after data collection; planned structure below]

5.1 Data Quality and Sample

[todo: final N per group; trials recorded and excluded; artifact percentage per condition (table); drops per movement trial]

5.2 Descriptive Statistics

[todo: table: mean HR, RMSSD, SDNN, pNN50 and α_1 per condition and expertise group, mean (SD)]

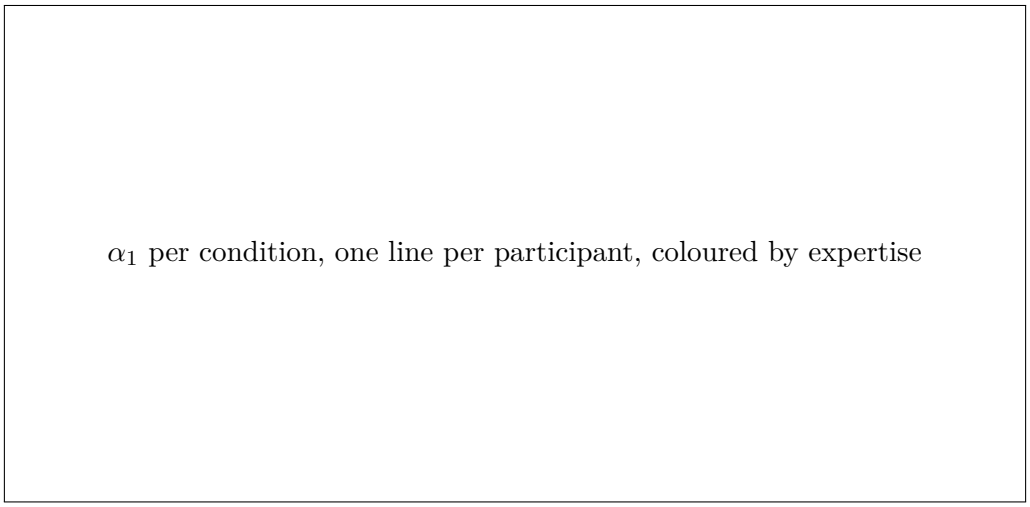


Figure 2: [todo: Figure 2]

5.3 Task versus Rest (H1)

[todo: model estimates; direction counts; position relative to 0.5, 0.75, 1.0]

5.4 Isolations versus Spins (H2)

[todo: model with mean HR; figure of $\Delta\alpha_1$ against ΔHR coloured by movement]

5.5 Expertise (H3)

[todo: group comparison and continuous analysis]

5.6 Exploratory Analyses

[todo: carry-over; hand; Bland–Altman of real-time vs offline α_1; questionnaire ratings]

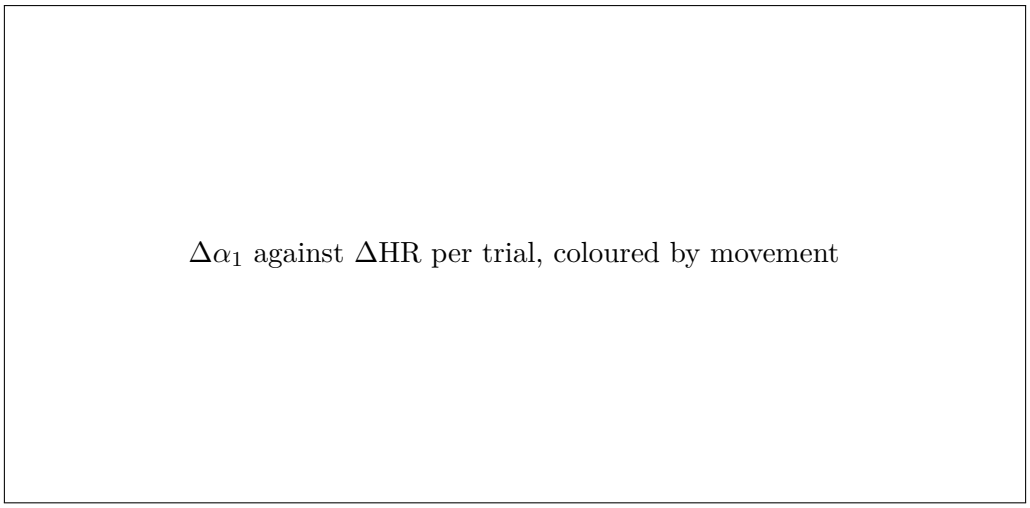


Figure 3: [todo: Figure 3]

6 Discussion

[todo: opening paragraph after results: restate the question, the main finding and its direction in three to four sentences]

The discussion is organised around three questions: in which direction α_1 moves when a person manipulates the stick, whether that movement is specific to skilled manipulation or explained by physical load, and what the answer implies for the stick as a tool for studying skilled engagement. The interpretation depends on the outcome, and Table 4 sets out in advance how each pattern would be read.

Table 4: Pre-specified interpretation of possible outcomes.

Outcome	Reading	Next step it motivates
α_1 falls towards 0.75–0.5, explained by HR	Ordinary intensity effect; the stick behaves like light exercise	Match conditions on HR before testing skill effects
α_1 falls beyond what HR predicts	Task-specific change in autonomic organisation, consistent with engaged regulation or co-activation [32]	Relate to flow ratings; compare isolations and spins at matched HR
α_1 stays near 1.0	Fractal organisation preserved under a demanding motor task	Compare with a non-skilled movement control
α_1 rises towards 1.5	Shift towards regularity, e.g. breathing locked to the movement rhythm	Record breathing; test respiratory entrainment directly
Expertise or hand differences	Skill demand modulates autonomic organisation	Longitudinal practice; movement quantification

6.1 Direction of the Shift

[todo: interpret the H1 result against the landmarks of Table 1]

If α_1 falls during manipulation, the most economical reading is the one established in exercise research: as physical demand rises, the many regulatory influences on the heart are progressively dominated by a single drive, and short-term correlations weaken [20, 21]. A fall towards 0.75 would

place stick manipulation in the range of light to moderate exercise, which would be informative in itself, because the movement looks physically undemanding. A fall towards 0.5 at only modestly raised heart rate would be harder to explain by load and would point to a change in the organisation of regulation.

If α_1 rises, the complexity framework offers a different reading. A shift towards 1.5 indicates that a slow, regular component has come to dominate the series [19]. In the present task the most likely source of such a component is breathing. If breathing locks to the rhythm of the movement, as locomotor–respiratory coupling would predict [2, 24], respiratory sinus arrhythmia imposes a regular oscillation on the heartbeat at the movement frequency or one of its subharmonics. A rise in α_1 would then be a cardiac trace of coupling between movement and breathing, which the present design cannot confirm without a respiratory signal but which can be tested directly in future work.

If α_1 stays near its resting value, the result is still informative. It would show that fractal organisation is preserved during a task that demands continuous visuomotor control, in contrast to endurance exercise of comparable heart rate, and would argue against the view that any engagement of the motor system reduces correlation structure.

6.2 Skill or Load?

[todo: interpret H2: does the isolation–spin difference survive the HR adjustment]

The heart-rate adjustment is the crux of the design. Two minutes of stick manipulation raise heart rate, and α_1 is known to fall with heart rate, so a raw difference between rest and movement is expected on physiological grounds alone. The question is whether isolations and spins, which differ in their demands on control and rhythm, differ in α_1 at matched heart rate. Isolations, which load visual prediction and fine control, are the more likely candidate for an effect beyond load; spins, which load rhythm and timing, are the more likely candidate for respiratory entrainment. [todo: relate the observed pattern to this contrast]

Two caveats apply. Heart rate is an imperfect proxy for physical load in upper-limb tasks, because small muscle groups raise heart rate less per unit of effort than whole-body exercise, and isometric holding in isolations may raise blood pressure more than heart rate. And the covariate adjustment assumes that the relation between heart rate and α_1 is the same in both movements, which can be checked but not guaranteed in a small sample.

6.3 Expertise

[todo: interpret H3]

The expertise hypothesis follows from the stages of skill learning [18]: if novices perform the movements under explicit control and experts largely automatically, the regulatory demand of the same movement differs between them. A smaller departure from resting α_1 in experts at matched heart rate would be consistent with this account and with the transient hypofrontality view of skilled performance [15]. The literature on rhythmic coupling adds a second possibility: if experts have learned to coordinate breathing with the movement, their α_1 could shift further towards regularity, as stable coupling imposes a regular respiratory rhythm. These two accounts predict different directions, and the observed pattern discriminates between them [todo: which one]. A cross-sectional comparison

shows whether such a difference exists. A longitudinal study is needed to show that practice causes it.

6.4 The Stick as a Research Tool

[todo: interpret the Bland–Altman result and data quality]

A secondary aim was to establish whether a low-cost chain of consumer hardware and open software can measure α_1 during object manipulation. Data quality during movement, the agreement between real-time and offline estimates, and the practicality of two-minute trials all bear on this. If agreement is poor, as the pilot recordings suggested it might be, future studies need to compute α_1 offline from raw RR intervals with documented settings, as recommended for exercise research [25].

6.5 Limitations and Future Research

Several limitations follow from the design. Trials lasted two minutes, which yields one estimate of α_1 per trial and no information about how α_1 develops within a movement bout; longer bouts are not feasible for isolations without repeated breaks. Upper-body movement introduces artifacts, and the chest strap's error for α_1 grows during activity and in the uncorrelated range [9, 38]. Breathing was not recorded, so respiratory changes during movement, a plausible driver of any shift, cannot be separated from other mechanisms. The analysis relies on a monofractal short-term index; multifractal and long-range structure were not assessed [19]. The sample is small and the expertise groups smaller still, so the between-group comparison has limited power and should be read as an estimate. Finally, the study contains no non-skilled movement control matched for effort, so the contrast between skill and load rests on the statistical adjustment for heart rate.

Each limitation points to a next step. A respiratory belt or the H10's accelerometer-derived breathing signal would allow coupling between movement, breathing and heart rate to be quantified directly, using the frequency-locking and phase-coupling measures of the coupling literature [23]. A metronome condition would test whether external rhythm stabilises the coordination, as it does in cycling [24]. A non-skilled rhythmic arm movement matched for heart rate would provide the control that the present design approximates statistically. And a longitudinal study, as originally proposed for this thesis, would test whether daily practice changes resting α_1 and the response to the movements, which is the question that the health relevance of fractal dynamics ultimately depends on [19, 31]. [todo: expand to the length of the other discussion subsections once results are in]

7 Conclusion

[todo: write after results, about half a page]

This thesis asked whether two minutes of rhythmic stick manipulation shift the fractal organisation of the heartbeat, as indexed by DFA α_1, and whether any shift is specific to skilled manipulation or explained by physical load. [todo: main finding and direction]. [todo: specificity after heart-rate adjustment]. [todo: expertise]. Beyond the specific result, the study shows how a portable recording chain built from a chest strap, a phone and open software can measure the organisation of cardiac regulation during skilled object manipulation, and it sets out the controls (heart rate as a covariate, breathing, a matched non-skilled movement) that future work on skill, flow and autonomic complexity

will need.

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8 Appendix

8.1 Abbreviations

BLE	Bluetooth Low Energy
bpm	beats per minute
DFA	detrended fluctuation analysis
α_1	short-term scaling exponent of DFA (box sizes 4–16 beats)
α_2	long-term scaling exponent of DFA (box sizes 16–64 beats)
ECG	electrocardiography
fBM	fractional Brownian motion
fGN	fractional Gaussian noise
HR	heart rate
HRV	heart rate variability
ICC	intraclass correlation coefficient
LoA	limits of agreement
LRC	locomotor–respiratory coupling
pNN50	percentage of successive RR differences larger than 50 ms
RMSSD	root mean square of successive differences
RR	interval between successive R waves (beat-to-beat interval)
SDNN	standard deviation of normal-to-normal intervals
VT1	first ventilatory threshold

8.2 Trial File Format

Each trial was saved as a JSON file with the fields listed below, alongside a line-delimited log of every raw message received from the streaming application.

`session_id`	condition, participant and start time
`condition`	condition label, e.g. `baseline`, `isol_l`, `spin_r`
`participant_id`	pseudonymous participant code
`recorded_at`	ISO timestamp of trial start
`target_duration_s`	set trial duration in seconds
`sample_count`	number of RR intervals received
`samples`	list of RR intervals with arrival timestamps
`computed_stream`	real-time metrics reported by the streaming application

8.3 Supplementary Tables

[todo: per-participant artifact rates; per-participant α_1 per condition; full model outputs]